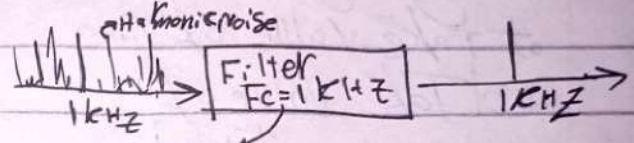
sheet(1)

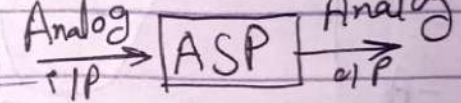
#1.1

- Signal → Physical quantity versus Time or Space or independent variable or variables.
  $x(t) = 20t$ $x(v, z) = 20v^2 + 5z$
  $(v \& z)$ → independent variable.
  $x(v, z)$ → dependent Variable.
- System → Physical device To Perform operation on Signal.

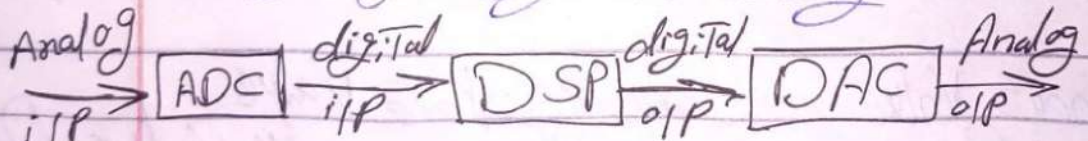


Reduce To noise.

- Signal Processing → Operations Perform to Signal
  There are 2 types:
  1. Analog Signal Processing

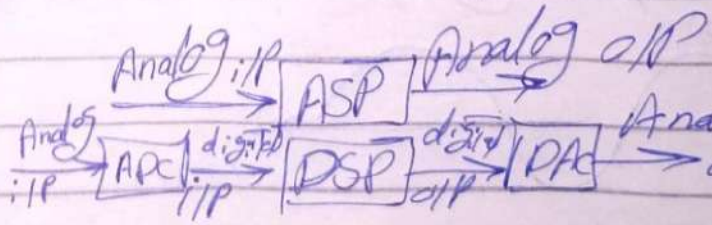


  2. Digital Signal Processing



#1.2

1. ASP
2. DSP



#1.3

* Multichannel
Signal generate from multi source.

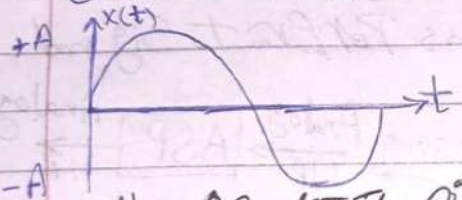
$$X(t)=\begin{bmatrix} X_1(t) \\ X_2(t) \\ X_3(t) \end{bmatrix}$$

* Multidimensions
Signal is function of signal independent variable.

$X(t) = 20t^2$

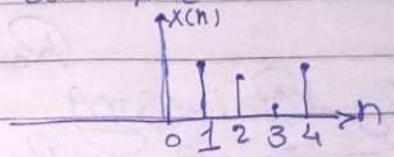
$X(V, Z) = 4V^2 + 43Z$

* Continous Time
Analog Signal $X_a(t)$ take value of Time and Take value of Continous interval.



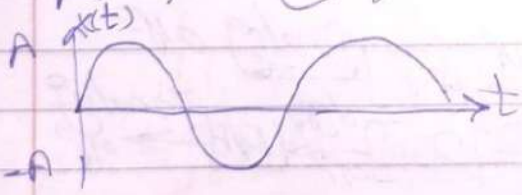
$X(t) = A\sin(2\pi Ft + \theta^\circ)$

* Discrete Time
Signal defination only to specific value at time.



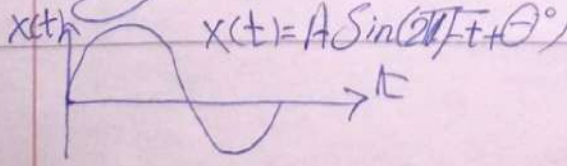
$X(n) = A\sin(2\pi Fn + \theta^\circ)$

* Continous Value
Take All Possible Value Finite or infinite.



* Discrete Value
Take one only value for Finite of Possible Value.

* Deterministic
Any Signal Can Be describe By mathematical Expression, Table of data and graph.



* Random
Any Signal Can't describe By mathematical Expression and Table of data, graph.

#1.4 A/D Converter

- Sampler → Convert Continous Time Signal into discrete Time Signal By Taken Sample of Continay Time Signal To Discrete Time instants.
- quantizer → Convert Discrete Time and Continay Value To Discrete Time and discrete value.
- Coding → in Coding Process each discrete Value is Represented By B-Bits Binary Squence.

#1.5

- Sampling Theorm → In Signal with max freq (f_{max}) Can Be Sampled By Sampling Rate (f_s) Condition greater than Twice max freq. [$F_s \geq 2F_{max}$].
- Nyquist Rate
  - ↳ Minimum Sampling Rate For Sample Signal without Aliasing.

#1.6 @ $x_a(t) = 3\cos(5t + \frac{\pi}{6})$

$x(t) = A\cos(2\pi Ft + \theta)$

$2\pi Ft = 5t \rightarrow F = \frac{5}{2\pi}$ $\therefore T_p = \frac{2\pi}{5}$ Sec → (Fundamental Period.)

$x_a(t + T_p) = 3\cos\left(5(t + \frac{2\pi}{5}) + \frac{\pi}{6}\right)$

$= 3\cos(5t + 2\pi + \frac{\pi}{6}) = 3\cos(5t + \frac{\pi}{6})$

$\therefore$ Signal is Periodic every $\frac{2\pi}{5}$ Sec.

(b) $X_a(t) = 3\cos(100\pi t)$

$x(t) = A\cos(2\pi F t + \theta)$

$2\pi F t = 100\pi t \rightarrow F = 50\,Hz \quad \therefore T_p = \frac{1}{50} = 20m\,Sec$

$x_a(t + T_p) = 3\cos(100\pi(t + T_p))$

$= 3\cos(100\pi t + (20m\,Sec \times 100\pi))$

$= 3\cos(100\pi t + 2\pi)$

$= 3\cos(100\pi t)$

$\therefore$ Signal is Periodic every 20m Sec.

(c) $x(n) = \cos\left[\pi\left(\frac{30n}{105}\right)\right]$

$x(n) = \cos\left(\frac{30n\pi}{105}\right)$

$x(n) = A\cos(2\pi F n + \theta)$

$2\pi F n = \frac{30 n \pi}{105} \rightarrow F = \frac{30}{210}\,Hz$

$\therefore F = \frac{K}{N_p} = \frac{30}{210} = \frac{1}{7}$

$x(n + N_p) = \cos\left(\frac{30\pi(n + N_p)}{105}\right)$

$= \cos\left(\frac{30\pi n + (30\pi \times 7)}{105}\right)$

$= \cos\left(\frac{30\pi n}{105} + 2\pi\right) = x(n)$

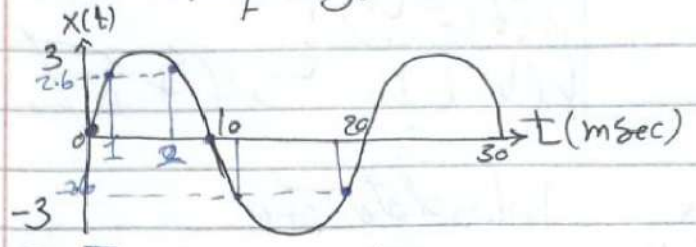
$\therefore$ Signal is Periodic every 7 Samples.

#1.7 $x_a(t) = 3\sin(100\pi t)$

(a) $x(t) = A\sin(2\pi F t + \theta)$

$2\pi F t = 100\pi t \rightarrow F = 50\,Hz$

$\therefore T_p = \frac{1}{F} = \frac{1}{50} = 20\,msec$



(b) $F_s = 300$ Sample/Sec

$F = \frac{F}{F_s} = \frac{50}{300} = \frac{1}{6} = \frac{1}{N_p}$ cycle/Sample.

$x(n) = 3\sin(2\pi F n + \theta)$

$= 3\sin(\frac{2\pi}{6} n) = 3\sin(\frac{\pi n}{3})$

$x(n+N_p) = 3\sin(\frac{\pi(n+N_p)}{3})$

$= 3\sin(\frac{\pi n}{3} + 2\pi)$

$= 3\sin(\frac{\pi n}{3}) = x(n)$

$\therefore$ Signal is Periodic every 6 Samples.

$x(n) = 3\sin(\frac{\pi n}{3})$

$x(0) = 0$ $\quad x(2) = 2.6$ $\quad x(4) = -2.6$

$x(1) = 2.6$ $\quad x(3) = 0$ $\quad x(5) = -2.6$

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(c) F_s?

~~$3\sin(\frac{\pi n}{3}) = 3$~~

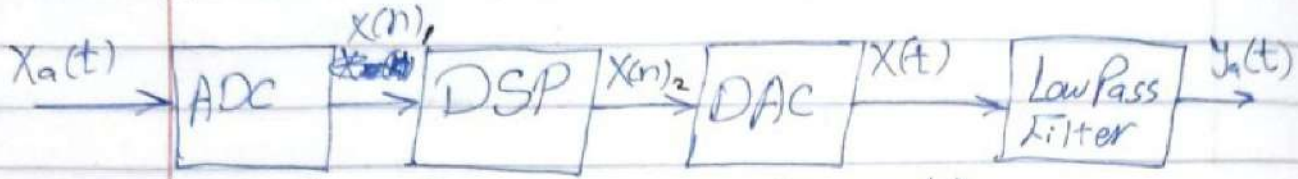
$x(n) = 3\sin(2\pi F n) = 3$

$= 3\sin(2\pi n \frac{F}{F_s}) = 3$

$3\sin(\frac{100\pi n}{F_s}) = 3$

$\frac{100\pi n}{F_s} = \frac{\pi}{2} \rightarrow$ $F_s = 200n$ $\quad$ $n=1 \rightarrow F_{min}$

#1.8 $T_1 = 5ms$ $T_2 = 1ms$



$x_a(t) = 3\cos(100\pi t) + 2\sin(250\pi t)$

$nT_s = \frac{n}{F_s} = \boxed{5\times10^{-3}n}$ $\quad t \rightarrow nT_s$

$x(n)_1 = 3\cos(100\pi \times 5\times10^{-3}n) + 2\sin(250\pi \times 5\times10^{-3}n)$

$= 3\cos(0.5\pi n) + 2\sin(1.25\pi n)$

$t \rightarrow nT_s \quad \therefore n = \frac{t}{T_s} \Rightarrow n = \frac{t}{1m\text{Sec}}$

$\therefore n = 1000t$

$x(t) = 3\cos(0.5\pi \times 1000t) + 2\sin(1.25\pi \times 1000t)$

$= 3\cos(\underline{500\pi t}) + 2\sin(\underline{1250\pi t})$

$F_1 = 250$ $\quad F_2 = 625$

Low Pass Filter
$y(t) = zero$

250 625

$F = 125$ F_1 F_2